

☒ EASY

☐ ALGEBRA

Algebra

10 questions · ~15 min

17

Q1**ALGEBRA**

If $4x - 28 = -24$, what is the value of $x - 7$?

A. -24

B. -22

C. -6

D. -1

A student council group is selling school posters for a fundraiser. They use the function $px = 5x - 220$ to determine their profit px , in dollars, for selling x school posters. In order to earn a profit of \$ 900, how many school posters must they sell?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$x = 10$$

$$y = x + 21$$

The solution to the given system of equations is x, y . What is the value of y ?

- A. 2.1
- B. 10
- C. 21
- D. 31

Q4

GRID-IN ALGEBRA

The equation $46 = 2a + 2b$ gives the relationship between the side lengths a and b of a certain parallelogram. If $a = 9$, what is the value of b ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$4x + 6 = 18$$

Which equation has the same solution as the given equation?

A. $4x = 108$

B. $4x = 24$

C. $4x = 12$

D. $4x = 3$

For the linear function f , the graph of $y = fx$ in the xy -plane has a slope of $\frac{1}{4}$ and passes through the point $(0, 5)$. Which equation defines f ?

A. $fx = \frac{1}{4}x + 5$

B. $fx = \frac{1}{4}x + \frac{1}{5}$

C. $f(x) = \frac{1}{4}x - \frac{5}{4}$

D. $fx = \frac{1}{4}x - 5$

Angela is playing a video game. In this game, players can score points only by collecting coins and stars. Each coin is worth c points, and each star is worth s points.

- The first time she played, Angela scored 700 points. She collected 20 coins and 10 stars.
- The second time she played, Angela scored 850 points. She collected 25 coins and 12 stars.

Which system of equations can be used to correctly determine the values of c and s ?

- A. $10c + 20s = 700$
 $12c + 25s = 850$
- B. $20c + 10s = 700$
 $25c + 12s = 850$
- C. $20c + 700s = 10$
 $25c + 850s = 12$
- D. $700c + 20s = 10$
 $850c + 25s = 12$

$$3a + 4b = 25$$

A shipping company charged a customer \$25 to ship some small boxes and some large boxes. The equation above represents the relationship between a , the number of small boxes, and b , the number of large boxes, the customer had shipped. If the customer had 3 small boxes shipped, how many large boxes were shipped?

- A. 3
- B. 4
- C. 5
- D. 6

Q9

GRID-IN

ALGEBRA

$$6x + k = 6x + 5$$

In the given equation, k is a constant. If the equation has infinitely many solutions, what is the value of k ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

A bus is traveling at a constant speed along a straight portion of road. The equation $d = 30t$ gives the distance d , in feet from a road marker, that the bus will be t seconds after passing the marker. How many feet from the marker will the bus be 2 seconds after passing the marker?

- A. 30
- B. 32
- C. 60
- D. 90